<u>Rise and Fall of Alviso's Historic Port: The Reuse, Repair, and Deconstruction of the Historic Town</u>

There is a quiet history that many Bay Area locals might not even know in San Jose at South San Francisco Bay, which has nothing to do with technology, startups, or venture capitals. However this one is about tides, salt flats, take back timber, and a small community that allows the stories of the town's history to continue circulating. When locals or visitors walk through the town of Alviso, it is like you are time traveling back into the past and getting to learn a little about how life was then. In the 19th century, Alviso used to be a busy port town for Santa Clara Valley, now Alviso has some remnants of historic buildings and industries, large salt ponds, and levees. There are three different transitions that have overlapped over two decades. They are tidal marsh restoration, rethinking of demolition and reusing, and grassroots efforts to repair the connection between the community and the environment. All of these efforts combined show a circular economy between the built and natural environment to build towards resilience in the town.

This article talks about the various industries that Alviso has held to think about ways to protect habitats and rethink waste like maritime, port tragedies, canneries, railroad, agriculture, and salt works. Alviso's story and history is not only local, but also global. It consists of climate risks, cultural history, economic connection, and how cleaning up waste can either make or break the health for future generations.

History of Alviso

Alviso, San Jose, California has a long standing history behind the town. Alviso is located at the South San Francisco Bay, next to Moffett Field and recently Topgolf. The town started off as home to the indigenous group called the Ohlone and later colonized by Spanish explorers which resulted in Pueblo de San José de Guadalupe and Mission Santa Clara de Asís being found in 1777. In 1838, Jaun Ignacio Alviso was given 6,353 acres of Embarcadero de Santa Clara, which was a part of the split up property area which then got settled in 1840.

In the mid 19th century, Alviso flourished with ships arriving and leaving Alviso to transport agricultural goods, tin, and produce to San Francisco Bay. Early Alviso had many adventures and industries like a shipping center, different canneries, and had steam boat accidents. As a result of these different industries, there were different communities that worked in each changing field creating a diverse community. There were Chinese, Japanese, Portuguese, and Mexican communities who lived in Alviso for various reasons.

Once the industries stopped operating, the buildings were all left behind as the Bay Area shifted to become a major technology hub. Alviso is now filled with a small residential community and recreation for walkers to the Don Edwards San Francisco Bay National Wildlife Refuge. However, some of the old buildings, levees, and abandoned structures are still standing throughout the town today. The City of San Jose recently documented the built heritage in Alviso in a Historic Context and Survey project, done to recognize the cultural area and whether to preserve or deconstruct the historic materials.

Alviso is home to not only humans and industry but also many of the species that are endangered or threatened. These consist of birds, mice, shrews, and plants as well as the sloughs and mudflats being important habitats for fish, shellfish, and infauna.

When driving into and around Alviso, people can see the leftover brick from the cannery, narrow streets that used to be wharves, and former salt ponds that reached the Bay. The area has been subsiding because of ground water extraction in the past resulting in Alviso being about 13 feet below sea level. With the addition of sea level rise and parts of Alviso lying at lower elevation has caused it to also be extremely vulnerable to flooding. There have been ten significant floods from the Guadalupe River and more minor scale floods throughout Alviso's history. This makes it difficult to know what should be protected, changed, or deconstructed since there are environmental factors at play as well.

Restoring the Bay Projects

South Bay Salt Pond Restoration Project

In the last twenty years, the only largest intervention that has occurred in Alviso is the South Bay Salt Pond Restoration Project. This was launched after state agencies, federal agencies, and support from private foundations acquired over 15,000 acres from Cargill of industrial salt ponds. This project is the biggest tidal wetland restoration project on the West Coast of the United States. The goals of the project are to restore tidal marshes and mudflats to protect wildlife, better flood protection, and recreation for Bay Area residents. The effort to turn industrial ponds to wetlands means removing and strengthening levees, letting water back, quantity of sediment and excavated materials, build trails, and make habitat features while having low carbon and waste impacts.

South San Francisco Bay Shoreline Project

Another project related to the South Bay Salt Pond Restoration Project is the South San Francisco Bay Shoreline Project. This project is a larger restoration project with focus on providing flood protection for the community in Alviso, better public access, habitat restoration in Alviso, and restoring some of the former salt ponds. The project aims to restore about 3,000 acres of former salt ponds and in order for the ponds to be restored, the flood protection has to be completed before. There is deconstruction being done for restoration efforts. This deconstruction is being done by changing the old levees to work for new marsh ecosystems.

Beneficial Reuse

The challenge during restoration work is figuring out whether extracted material can be used for other projects instead of being wasted, this is called beneficial use. San Jose has tried to incorporate beneficial reuse by using excavated material into raising the bottom of ponds or a type of support system for a wall so tidal marshes can re-establish themselves.
The Valley Transportation Authority's Beneficial Reuse Project has used excavated materials from infrastructure projects to speed up marsh restoration and habitat restoration.

There is both an aesthetic and ethical logic behind these approaches. Society has thought of natural materials that have been disturbed by humans, whether intentionally or unintentionally, as waste that can not be used for any purpose. Beneficial reuse makes it so that the materials found are local resources. This helps in multiple ways like reducing the emissions from transportation, reducing disposal costs, and letting people keep their jobs and value of the materials in the area. However, while working with the sediments there needs to be attention put towards whether there is any contamination in it, permitting problems, and coming to an agreement with agencies. Alviso shows and sets an example of other areas that ecological restoration can help to reduce waste and increase the value as well as health for people and the ecosystems.

Demolition vs Deconstruction

San Jose has shown more attention to deconstruction when it comes to building rather than demolition in recent years. They do this by carefully dismantling buildings to use beams, doors, bricks, and hardware for the new project. This is being done because demolition ends up destroying tons of reusable material to the landfill. With Alviso being a historic town, there are structures from the 19th and 20th century that have old timers and masonry that are irreplaceable. Deconstruction helps to preserve those materials and keep the local history strong by educating people on what was done to house the plethora of industries in the small area.

San Jose Construction and Demolition Diversion Program has implemented deconstruction of sites in 2001. This was the city's first efforts to reduce landfill waste that came from construction projects by incentivizing workers and contractors to recycle or reuse the materials that they are trying to demolish. The program has helped to shine light on and encourage reclaimed materials. Furthermore, Santa Clara County has promoted green salvage to help recover more materials and donate items that were found.

Deconstruction also impacts that budget and the time needed for the project to be completed. It needs more workers, space for storage or resale of the materials, and people willing to buy products that might not be as strong as newer, unused materials. This means buyers would willingly purchase imperfect and aged woodland hardware. On the brighter side, the effects on the environment are extremely positive and significant. This is because there is less virgin materials needed, organizations can purchase reclaimed materials at a lower cost for community repairs, and the carbon footprint is lower since the carbon in the reused items is already in it. For

Alviso, this allows for the local stories in the buildings and the town history to be used in other areas which help the history live on.

Community and Preservation

Material reuse and restoration goes further to those whom this benefits. Alviso is the focus for City of San Jose's Alviso Historic Context and Survey Project along with other planning processes that work to document and preserve the town's cultural heritage. These processes matter because there is also a social aspect the the reusing of materials not just to build with. The community, City, and the deconstruction company need to decide which building needs to be preserved, which should be deconstructed, what is usable, who can use the materials, and many more questions that need to be answered.

An example of this is a partially burned building, the historic H.G. Wade Warehouse, that was approved for demolition by San Jose officials after intensive discussion. This decision came to be after the site was neglected and then a fire in 2021 that destroyed most of the few standing walls of the building. This warehouse was also the last trace of a Civil War era building. The commissioner added that any bricks from the warehouse that were removed had to be received and protected for future use. So although this is labeled as a demolition of the burned warehouse, it is actually a deconstruction since the bricks were saved for future development on the site.

Local organizations and other park projects have made it their mission to voice the opinions of the community on preserving the historic sites in Alviso. There have been improvements to Alviso Park by adding in public amenities by acknowledging Alviso's cultural heritage and ecological environment. It is important to include the community in the decision making for their town. By including them in the discussion, it can help residents to understand the deconstruction and restoration efforts that are being done to recover materials from historic sites which can be used again within the community. An example of this would be using timber from the deconstruction of the historic building to be used in playgrounds, building a community garden with materials, or bricks being used in new constructions in Alviso.

Waste, Recycling, and Infrastructures

Alviso's story goes beyond the history, but also shows urban waste systems. Although there is beneficial reuse and salvage occurring on the industry and building side, households and small businesses in Alviso still need recycling and waste disposal for everyday life. The Bay Area has many recycling along with construction and demolition facilities that process that waste. There are also private recycling groups like GreenWaste and other centers that locally deal with compost, recycling, construction recycling, and other waste in the area. Without processing or buyers close by for the reclaimed materials, it makes it hard for there to be more salvaging or excavation work of material found on sites.

A winding recreational trail with no trees and a lot of foliage surrounds the salt flats on the edge of the town of Alviso. There is a system in place where on certain days, incarcerated individuals are transported to Alviso by bus accompanied by correctional officers, who spend hours collecting litter on the trail pathway. During one visit, there were bags of trash gathered in neat piles beside the trail as the group of incarcerated individuals who are correctional officers were on a lunch break. There were more than just bags that were gathered. There were objects like a tire, large canisters, many plastic bottles, and balls.

Who Works for Salvaged Materials?

Salvage and reused materials can either reinforce the inequalities or help to remove that idea. There are many different aspects that need to be considered when getting reclaimed materials, like who gets paid, who works, and who profits. In Alviso, the rest of the Bay Area and anywhere else in the world, this can be both a risk or an opportunity. There are also different rules and regulations set in place which differ from place to place. However, from the city in which Alviso is located, there are policies that make the profits from the reused materials back to the community programs to help shift the more efforts toward justice and equity in the labor sector.

There can also be an aspect of informal labor practices where waste picking and salvaging is predominantly informal workers who put their lives at risk. In the Bay Area however, this is not seen or noticed as much since they are smaller scale or independent workers. Creating fair wages and safer working conditions can help for more circular, ethical, and durable working practices.

Circular Flood Resilience

Sea level rise in Alviso means that levees have to be raised, which requires stronger infrastructure and, in turn, restructured floodplain plans. The restoration projects in the South Bay show the use of excavated materials used to rebuild the pond floors and make marshes. This cuts the carbon footprint that would have been created from the raw materials. With the mitigation of raw materials, it allows for the ecological benefits to show results sooner. Agencies need to test sediment quality so that there are no contaminates in the mix and the permits for reuse are carefully checked.

Adaptation is not only changes made for the environment, but also an adjustment for the culture of the society. Alviso residents need to find safety or elevated housing. Something that can be done for Alviso residents in flood prone areas is to use reclaimed materials to build a sort of flood barrier around their home or put their house on stilts for residents to continue living in Alviso without flood risks. Other salvaged materials can also be used like reclaimed sand or compacted fill from a local source to raise protection efforts from floods. Much of the materials needed can come from the port area, so closer to the Bay or salt flats to build up flood resilience. The idea is to plan early for deconstruction and salvaged materials to provide resilient infrastructures as part of restoration efforts.

Reuse: Cannery Bricks to the Community

One of the most impactful parts of reusing materials is the memory of the object continuing to live on. Bricks from the old cannery can be made into walls for new structures like a community garden or timber from the houses workers use to live in can become benches on the trails with added writing to inform trail walkers about the town's history. These small creations can preserve the value the building had in the past while satisfying the needs of the town in the present.Local artists and crafts people have been helping to realize the possibilities of using reclaimed materials as well as their stories to make new objects.

Some community groups, outside of Alviso, have done projects like this. They create things like furniture, light fixtures, and artwork from the salvaged materials. An example of this is a Bay Area company called Building Resources which is based in San Francisco. They work to reduce solid waste by distributing salvaged building materials for reuse. These efforts are important to combine local history with the public.

Barriers and Circular Economy

Deconstruction costs more than demolition because of the need for more labor and time needed to take the structure apart along with figuring out what can be reused and what can not be reused. Materials that are salvaged need a proper place with the right conditions to be stored in and a market where people can buy these materials if they want. Sediments being beneficial reused need to be tested for contamination and need to have the proper permitting for it to even take place. Thankfully, the market for reclaimed materials has grown in value because of the aesthetic and people understanding the environmental reasons of why it is important. Municipalities are making the construction and demolition diversions have stricter goals. There are grant programs specially for climate adaptation and conservation of site heritage.

Materials and Memories

The salvaged timber and brick are not just supplies to be reused for a more circular and sustainable economy, but they carry the memories and history that the town has held on to. The resilience and circularity that Alviso, as well as probably other towns, has shown preserves that care of the ecology and the story. From testing and making sure the sediments can be reused while also commissioning artwork in the community from the reclaimed materials to salvaging storefronts to write about them in books and library pages to help remember the historic port town of Alviso.

The future of Alviso is shaped by the choices of what people decide is worth keeping and what will be thrown away. The slough meeting the Bay has a somewhat significant meaning behind it like when you visit an art museum and look for the meaning behind the art works. The slough and the Bay meet at a very narrow piece of land. Just like the strength Alviso has had, it represents the resilience of new technology and value that can be found. Reusing shows that value in the history left behind. Repairing restores the usability of the materials before they are completely unusable and the story is gone. Deconstruction allows for opportunity to be found in something one person might have seen as waste. Together, they work to reincarnate the beauty and history that could have been lost forever.

See if somewhere in your home town there can be changes made to create a more circular and sustainable economy by using materials from building or excavated materials to build new buildings.